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Histone Variants as Stem Cell Biomarkers for Long-Term Injection Medialization Laryngoplasty

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Objectives/Hypothesis: Vocal fold (VF) paralysis by sectioning the recurrent laryngeal nerve dramatically impacts the life of thyroidectomy patients. Volume-expanding materials can temporarily restore VF medialization. To prolong this benefit, adipose mesenchymal stem cells (ADSCs) and micronized acellular dermis (MACD) were co-injected in a rabbit model of injection medialization laryngoplasty. Biomarkers of in situ proliferation were identified by mass spectrometry proteomics and pathway analysis to guide future efforts to increase the length of benefit.

Methods: ADSCs were expanded and/or differentiated into chondrocytes (CHON) as collagen microspheres. After VF paralysis rabbits received MACD, MACD + undifferentiated ADSC, or MACD+CHON, ADSCs differentiated into chondrocytes. After 12 weeks, animals were sacrificed and 5-μm paraffin-embedded cryosections were prepared from larynges for hematoxylin and eosin visualization and nanoflow liquid chromatography electrospray-ionization tandem-mass spectrometry analysis of tissue collections. Validated proteins were processed by Venn subtraction and gene ontology (GO) overrepresentation analysis to identify unique pathways and biomarkers.

Results: Confirmed proteins numbered 147 (MACD), 1,243 (MACD+ADSC), and 1,033 (MACD+CHON). Totally, 333 proteins were uniquely found in the MACD+ADSC group, including mesenchymal surface markers CD9, CD44, fibronectin, and vimentin. Over 70% of proteins belonged to catalytic activity and binding GO categories, with the histone (H) family being overrepresented ($P < 0.05$). Histone variants H3.3, H2A.V, and H2A.Z (associated with open chromatin states) were overrepresented in the MACD+ADSC group, whereas structural histones H2A, H2B, and H4 were not.

Conclusion: Biomarkers, including atypical histones, are associated with in vivo proliferation of ADSCs and an expanded VF medialization volume.

Key Words: Injection Laryngoplasty, Adipose Derived Stem Cells, Atypical Histones.

Level of Evidence: NA

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INTRODUCTION

Stem cells are fast gaining popularity in tissue repair applications, but issues relating to establishing standardized practices persist. Our laboratory is gaining expertise with new biologicals to address vocal fold (VF) paralysis that is caused by accidental sectioning of the recurrent laryngeal nerve (RLN) during thyroidectomy. Thyroidectomy performed for the management of thyroid carcinoma, Graves disease, and multinodular goiter is increasing in popularity.[1] Over a 15-year period (1995–2009), a 94% increase in the procedure was reported. The trend continues unabated, in keeping with the growing incidence of thyroid cancer over the last three decades.[2]

Currently, over 130 thousand thyroidectomies are performed each year,[3] bringing to the forefront the issue of maintaining RLN integrity during surgery. Unfortunately, VF paralysis persists with a ~2% incidence rate, even with guidelines in place.[4] Because proper closing of the upper pharynx is necessary for lung and laryngeal integrity, a common practice to restore VF medialization is to inject volume-augmenting materials that bring the paralyzed VF proximal to its counterpart. This procedure is called *injection medialization laryngoplasty* (IML) and can be performed uni- or bilaterally, depending on RLN integrity. When successful, IML can provide upward of 1 year of clinical benefit, but results vary according to the materials used.[5,6] Currently, there is great interest to further extend IML benefits using new reagents such as stem cell preparations. To advance this research, it is imperative to monitor the fate of injected mesenchymal stem cells in vivo. However, this task is challenging, partly because injury sites become populated with stem cells mobilized from other body compartments. To distinguish exogenous from endogenous cells, injected stem cells have been labeled, but dyes and biologicals are rapidly extruded and the signal fades within days. We previously showed that a combination of micronized acellular dermis (MACD)

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and adipose mesenchymal stem cells (ADSCs) increased long-term IML medialization volume.[7,8] We hypothesized that continued in vivo proliferation of ADSCs was mostly responsible for these results, but inflammatory processes may have also played a part. Because ADSC labeling procedures are unreliable for long-term use, we designed a protocol to monitor biomarkers of ADSC proliferation using nanoflow liquid chromatography electrospray-ionization tandem-mass spectrometry (nLC-MS/MS) proteomics analysis.

Here we describe a nLC-MS/MS protocol that was successfully used with scrapped material recovered from fixed, hematoxylin and eosin (H&E)-stained tissue sections to identify the ADSC biomarkers CD9, CD44, fibronectin, and vimentin—as well as the histone (H) variants H2A.V, H2A.Z, and H3.3 in rabbits at 12 weeks following an IML injection with MACD+ADSC.

MATERIALS AND METHODS

Sample Preparation

All animal experiments were performed in agreement with the Guide for the Care and Use of Laboratory Animals and the U.S. Public Health Service's Policy on Humane Care and Use of Laboratory Animals. Procedures for animal surgery, IML, tissue fixation and embedding, sectioning, and staining were previously described.[7] Briefly, 12 weeks post-IML rabbits were sacrificed, their larynges collected and fixed in 10% formaldehyde for 24 hours or more. Next, the tissues were paraffin-embedded and sectioned such that comparable depths of the paraffin blocks could be queried across groups. To achieve this, three 5-μm thick slices were cut every 400 μm, spanning the entire length of the blocks. This resulted in approximately 27 to 39 tissue sections, of which equal numbers were stained with H&E or Alcian Blue (for chondrogenic differentiation) and photographed to compile a digital record. Next, the cover slips on the H&E slides were removed by overnight incubation with xylazine, and then they were extensively washed (10×) and air-dried. To maintain uniformity during tissue collection, an 11-mm section that extended caudally from the cricoid cartilage and contained the IML ellipsoid was considered (Fig. 1A). Ellipsoids were scraped off with disposable scalpels onto weighing paper, pooled into 0.5-mL polypropylene microfuge tubes, and taken to the Medical Genome Facility-Proteomics at the Mayo Clinic, Rochester, MN, for protein analysis by nLC-MS/MS. Two independent runs were performed per group, each consisting of tissue samples from two pooled animals.

Peptide Identification by nLC-MS/MS Analysis

Tissue scrapings were solubilized in 50 μl of 2% Sodium dodecylsulfate (SDS)/50 mM Tris (pH 8.2) at 98°C for 1 hour and reduced with 10 mM Tris-(carboxyethyl) phosphine hydrochloride (TCHP) at 56°C for 30 minutes. Cysteines were alkylated with 10-mM iodoacetemide at room temperature for 30 minutes in the dark to prevent adventitious alkylation. Following SDS removal with Pierce Detergent Removal Spin Column, samples were digested at 37°C overnight with 1:50 trypsin solution, and acidified digests were analyzed by nLC-MS/MS using a Thermo Scientific Q-Exactive Mass Spectrometer coupled to a Thermo Ultimate 3000 RSLC nanoflow high-performance liquid chromatography system (Thermo Fisher Scientific, Bremen, Germany). Digested peptides were loaded onto a 250 Nanoliter (nl) OPTI-PAK trap column (Optimize Technologies, Oregon City, OR) custom packed with Michrom Magic C8 solid phase material (Michrom Bioresources, Auburn, CA), and chromatography was performed with 0.2 % formic acid in both A solvent (98% water/2% acetonitrile) and B solvents (80% acetonitrile/10% isopropanol/10% water), with a 5% A to 40% B gradient over 80 minutes and running at 400 nl/minutes through a PicoFrit 100 μm × 350 mm column (New Objective, Woburn, MA),

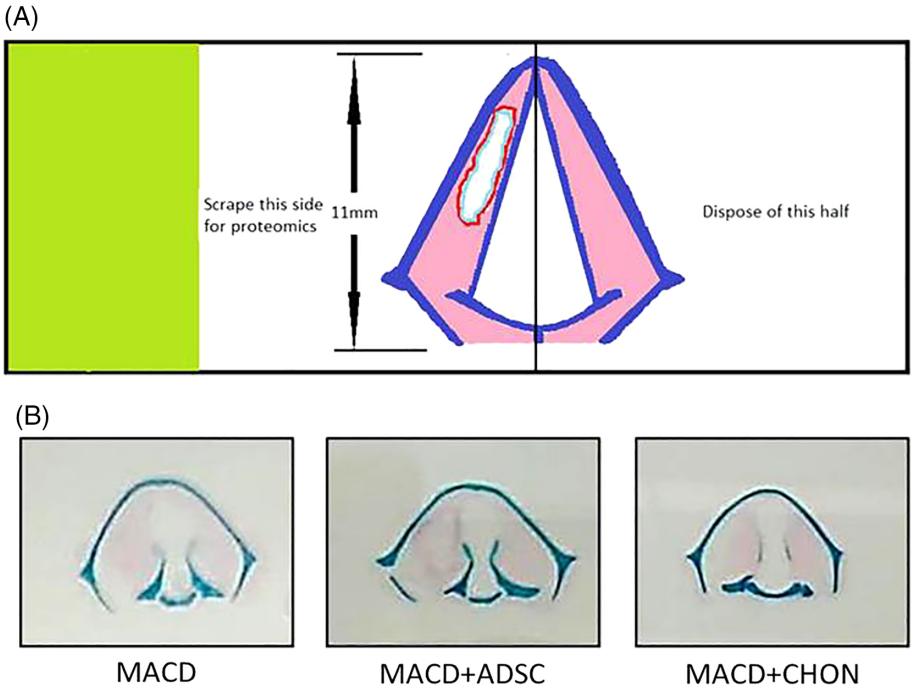


Fig. 1. (A) Diagram of the slide-scraping. collection protocol. (B) Representative tissue sections from paraffin blocks after hematoxylin and eosin staining showing the injection medialization laryngoplasty ellipsoid on the injected (left) side. ADSC = adipose mesenchymal stem cell; CHON = chondrocyte(s); MACD = micronized acellular dermis.

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hand packed with Agilent Poroshell 120 EC C18 stationary phase (Agilent, Santa Clara, CA). The mass spectrometer setup was a Fourier Transform full scan from 340 to 1500 mass/charge ratio (m/z) at resolution 70,000 (at 200 m/z), followed by High-energy Collision Dissociation (HCD) MS/MS scans on the top 15 ions at resolution 17,500. The MS 1 automatic gain control target was set to $1e^6$, and the MS 2 target was set to $1e^5$ with maximum ion injection times of 50 ms and 50 ms, respectively. Dynamic exclusion places selected ions on an exclusion list for 30 seconds.

Database Searching

Tandem MS/MS peptide spectra were extracted using msConvert version 3.0.4019 (ProteoWizard) software and further analyzed with Mascot version 2.4.0 (Matrix Science, London, UK), and X! Tandem (thegpm.org/tandem) in order to match mass spectra with peptide sequences to identify, characterize, and quantitate proteins. The search was performed across the rabbit RefSeq protein database (12/2016 with 77,162 entries), which included a decoy reverse database, with trypsin as default. Searches were performed with fragment ion mass tolerance of 0.020 Da and parent ion tolerance of 10.0 PPM. Glu->pyro-Glu of the n-terminus, ammonia-loss of the n-terminus, and gln->pyro-Glu of the n-terminus were specified in X! Tandem as variable modifications. Oxidation of methionine and the iodoacetamide derivative of cysteine were specified as variable modifications in both Mascot (Matrix Science) and X! Tandem.

Criteria for Protein Identification

Scaffold version 4.6.2 (Proteome Software Inc., Portland, OR) was used to validate MS/MS-based peptide and proteins. Peptide Identifiers (IDs) were accepted if they could be established at greater than 95% probability by the Scaffold Local False discovery rate (FDR) algorithm. Protein IDs were accepted if they could be established at greater than 95% probability using Protein Prophet.[9] Only proteins containing two or more unique peptides were compared across groups.[10]

Bioinformatics

Venn subtraction analysis (bioinfogp.cnb.csic.es/tools/venny/index2.0.2.html) was performed using Venny 2.0.2 and Gene Ontology (GO) enrichment analysis using gProfiler (biit.cs.ut.ee/gprofiler/index.cgi) and Panther, with access from the GO Consortium GUI (www.geneontology.org/page/go-enrichment-analysis).

RESULTS

Representative laryngeal sections at 12 weeks post-IML are shown in Figure 1B. A grayish ellipsoid containing cells embedded into the MACD matrix is visible by H&E staining on the injected side but is absent from the noninjected, contralateral side. To identify unique proteins associated with the in vivo proliferation of injected cells, ellipsoids were scrapped off and processed for nLC-MS/MS analysis.

An initial set of validating experiments (not shown) was performed to compare the ADSC protein yield with results from the literature. Typically, ~3,000 proteins were identified for rabbit ADSCs in tissue culture, in agreement with previously reported data for human ADSCs.[11] Furthermore, a similar number of proteins with > two unique peptides was identified in tissue culture and excised laryngeal ellipsoids at 12 weeks post-IML ADSC injection (tissue culture: 1,883 ± 279; rabbit larynges: 1,327 ± 204).

Figure 2 shows results from one of two independent nLC-MS/MS runs, in which the number of unique peptides is plotted relative to the total number of identified proteins as well as the overlap between the two runs. In all cases, a mean value of ~3.0 peptide residues per protein was observed, with ~50% of all proteins identified by two or more unique peptides. Over 80% of proteins were present in both runs, consistent with previously published data.[12] For the illustrated runs, 456, 3,321, and 2,970 proteins were identified, respectively, in the MACD, MACD+ADSC, and MACD-CHON (chondrocytes[s]) groups. Of these, 147 (MACD), 1,243 (MACD+ADSC), and 1,033 (MACD+CHON (chondrocytes[s])) proteins were validated by two or more unique peptides. We hypothesized that proliferation of injected ADSCs accounted primarily for the

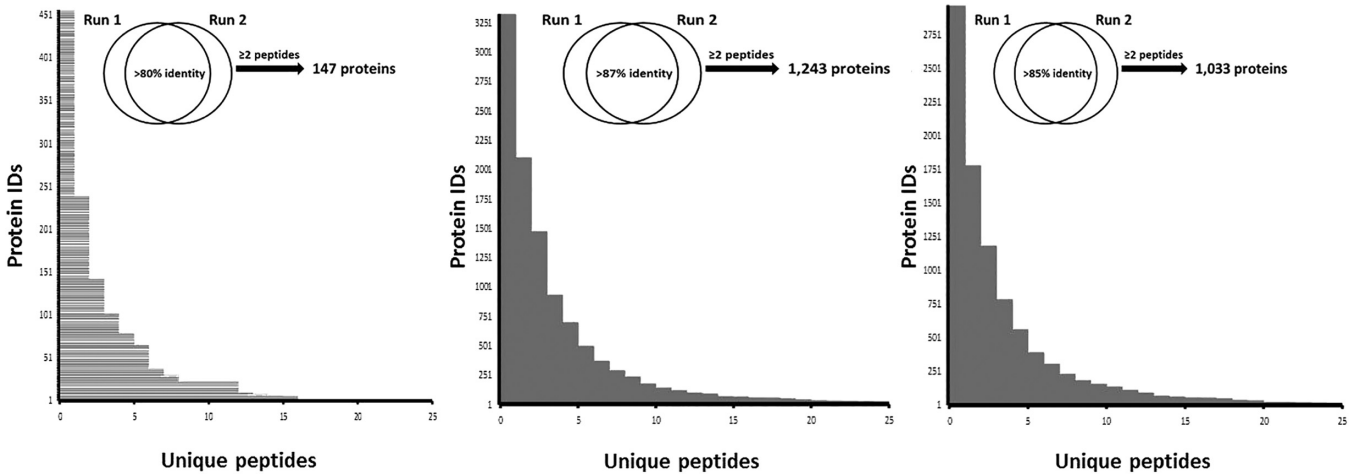


Fig. 2. Number of identified proteins in relation to their unique peptides. Proteins recovered from hematoxylin and eosin stained tissue sections were digested with trypsin and constituent peptides were identified by nLC-MS/MS analysis and database searches. Insert: work flow diagram. Common proteins between two independent nLC-MS/MS runs were validated by two or more unique peptides and processed for further analysis.nLC-MS/MS = nanoflow liquid chromatography electrospray-ionization tandem-mass spectrometry.

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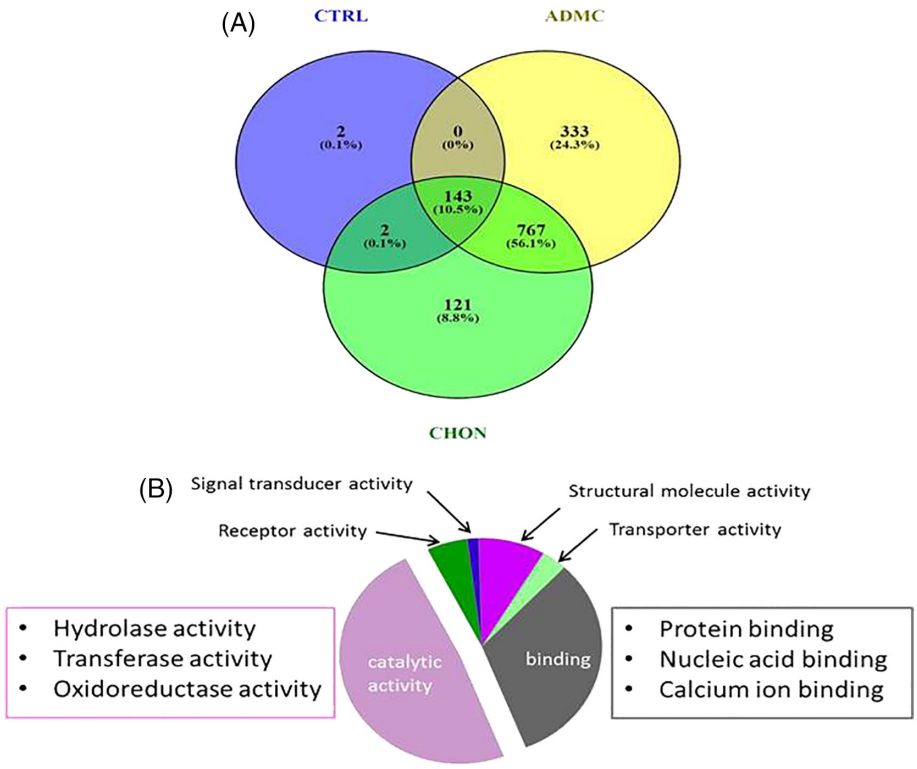


Fig. 3. (A) Venn subtraction diagram showing common and unique proteins between the three groups. (B) Pie chart of GO overrepresentation analysis results showing the principal GO categories with top three leading functional components. ADMC = Adipose Derived Mesenchymal Cells; CHON = chondrocyte(s); CTRL = Control; GO = gene ontology.

volume difference previously reported[7] and sought to identify proteins uniquely present in the MACD+ADSC group. A total of 333 unique proteins were identified by Venn subtraction analysis in this group (Fig. 3A). Notably, the adipose tissue-derived stromal cell biomarkers CD9, CD44, fibronectin, and vimentin[13] were among these proteins. To reveal the significance of the 333 proteins identified in the MACD +ADSC only, we performed GO over-representation analysis with a $P < 0.05$ level of statistical significance (Fig. 3B). Most proteins fell into two principal GO categories: catalytic activity and binding. The top five overrepresented processes in each GO category are shown in Table I. Within catalytic activity and binding GO categories, histones were overrepresented relative to other proteins. Because histones are essential for unwinding the DNA double helix to allow new DNA and RNA synthesis and enable cellular proliferation, we next examined the types of histone proteins involved. We found that in the MACD+ADSC group, the atypical histones H3.3, H2A.V, and H2A.Z are overrepresented relative to the other two groups. These histones have been implicated in processes associated with growth and proliferation.[14] On the other hand, histones H2A, H2B, and H4 that are associated with inactive nucleosomes are similarly represented in all three groups (Fig. 4).

DISCUSSION

Stem cells are fast gaining popularity in regenerative medicine due to their unparalleled ability to replenish cellular populations depleted by injury, disease, or radiation.[15,16] Here, we describe our results with ADSCs in a rabbit model of IML in which the goal was to induce long-term remedialization of the paralyzed VF.

IML is gaining popularity over framework surgery, with a reported fourfold surge in the number of procedures performed between 2003 to 2008.[17] The ease and relative safety of IML are important considerations. In contrast to framework surgery, IML does not require general anesthesia, can be performed on an outpatient basis, has fewer comorbidities, and costs about half as much to perform.[17,18] A wide assortment of injectable biomaterials have been used either for short-term correction: bovine gelatin (Gelfoam, Pfizer, New York, NY); carboxymethylcellulose (Radiesse Voice Gel [Merz, Raleigh, NC]); collagen (Cymetra [LifeCell Corp. Branchburg, NJ], Zyplast [Allergan Inc., Irvine, CA], Zyderm [Allergan Inc., Irvine, CA]); and hyaluronic acid (Restylane, Galderma Laboratories, Ft. Worth, TX) or longer lasting benefits (Calcium Hydroxylapatite, fat, fascia, polydimethylsiloxane (Bioplastique [Bioplasty, St. Paul, MN])). Overall, these biomaterials are stiffer relative to the natural viscoelasticity of the VF, giving rise to phonation issues. Because adipose tissue and the VFs have similar viscoelasticity coefficients,[19] we reasoned that proliferating adipose-derived mesenchymal stem cells could be a suitable material for IML, and at the same time present phonation correction advantages. Mesenchymal stem cells have been shown to proliferate in the heart,[20] bladder,[21] lung, intestine, and other organs,[22] and to promote tissue repair. We were attracted by the notion that a pool of ADSCs implanted next to the paralyzed VF would continue to proliferate and thus provide long-term medialization benefits.

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TABLE I.
Top Five Overrepresented Cellular Processes Identified by GO Overrepresentation Analysis at $P < 0.05$ Significance Level

Group	Process	*P* Value
MACD	Chromatin organization	3.91E-13
	Chromatin assembly	2.02E-12
	Muscle contraction	6.52E-09
	Glycolysis	4.1E-07
	Regulation of gene expression, epigenetic	1.6E-06
MACD+ADSC	Muscle contraction	2.64E-14
	Phosphate-containing compound metabolic process	1.27E-12
	Translation	4.54E-10
	Chromatin organization	4.74E-09
	mRNA splicing, via spliceosome	2.22E-07
MACD+CHON	Muscle contraction	1.13E-15
	Translation	3.94E-09
	Chromatin assembly	1.33E-07
	Phosphate-containing compound metabolic process	1.83E-07
	Intracellular protein transport	4.42E-07

ADSC = adipose mesenchymal stem cell; CHON = chondrocyte(s); MACD = micronized acellular dermis.

Previous studies have demonstrated that scaffolds increase cell survival by inhibiting apoptosis and upregulating stem cell biomarkers (Klf4, Sox2, and Oct4).[20,23,24] This strategy seems to have been met with success in our experiments. As shown above, at 12 weeks postinjection, cells remaining inside the scaffold included a population expressing ADSC and proliferation markers. In addition, vimentin, which is a stem cell biomarker, was upregulated > threefold in these cells (not shown).

In previous work, we tested the efficacy of a MACD +ADSCs bioreagent for promoting VF medialization post RLN sectioning. We noted that compared to MACD alone, a medialization promoting volume three times larger was observed in the MACD+ADSC group at 12 weeks postinjection.[7,8] Here, we focus on protein biomarkers of ADSC proliferation associated with the increased ellipsoid volume that correlate with the duration of clinical benefit. To achieve this, we first subtracted from the set of MACD +ADSC proteins the set of proteins expressed in the MACD group. We reasoned that the MACD group contained local proteins as well as proteins potentially expressed by endogenous, homing mesenchymal cells that are known to migrate to sites of injury, as well as proteins associated with postinjection tissue damage such as inflammatory factors secreted by infiltrating lymphocytes.[25] Second, we further subtracted from the MACD +ADSC protein set proteins expressed in the MACD +CHON group. By predifferentiating in vitro ADSCs into chondrocytes, we sought to generate a "decoy" cell population depleted of proliferation biomarkers expressed by undifferentiated ADSCs. The remaining set of proteins that is unique to ADSCs represents processes associated with in vivo proliferation. In an attempt to eliminate spurious results, we only considered proteins that were identified by two or more unique peptides and were common between two replicate runs, with each run consisting of pooled tissue from two animals.

The MACD+ADSC group yielded the largest set of proteins, consistent with the proliferative expansion of undifferentiated ADSCs, whereas the MACD+CHON group yielded relatively fewer proteins, in keeping with the more modest proliferation of phenotypically differentiated chondrocytes. Finally, the MACD group, in which no cells were injected, yielded the fewest proteins in this analysis. When subtraction analysis was performed, a total of 333 unique proteins were expressed in the MACD +ADSC group alone. Expression of the adipose tissue-derived stromal cell markers CD9, CD44, fibronectin, and vimentin was observed and used to confirm the ADSC nature of this cell population after 12 weeks' proliferation in rabbit larynges. The two largest, overrepresented GO categories were "catalytic activity" and "binding." In both categories, histones were overrepresented relative to other proteins. Upon further analysis, we found an overrepresentation of the atypical histones H3.3, H2A.V, and H2A.Z in MACD+ADSC group relative to MACD alone. These histones are associated with uncondensed chromatin and cellular proliferation.[14] In contrast, histones H2A, H2B, and H4, which are associated with condensed nucleosomes and quiescence, were equally represented in all groups. These initial results suggest that useful data may be obtained from fixed and H&E stained histopathology slides of animal tissues using mass spectrometry analysis. Such studies can help identify factors associated with

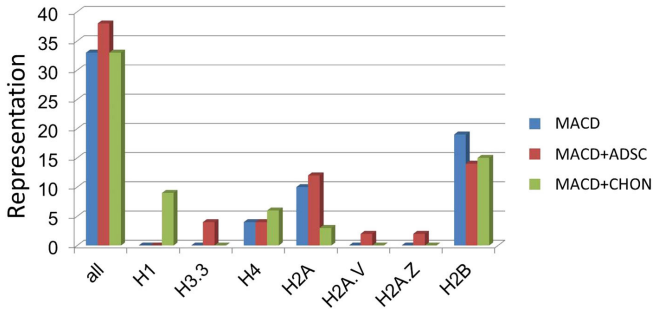


Fig. 4. Histone distribution across groups. ADSC = adipose mesenchymal stem cell; CHON = chondrocyte(s); H = histone; MACD = micronized acellular dermis.

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in vivo proliferation of ADSCs. This knowledge is useful to monitor the success of corrective interventions for VF paralysis in humans using stem cell preparations and biomaterials. Monitoring the fate of injected mesenchymal stem cells in vivo is challenging, in part due to contamination with endogenous mesenchymal stem cells that migrate to sites of injury. For this reason, injected cells are often labelled with dyes or biological reagents, such as SV40 T antigen DNA.[26] However, plasmids and dyes are readily excluded from stem cells by members of the ATP-binding cassette drug transporter family,[27] which limits the length of observation to a few days. One useful procedure has been to inject male MSCs into female recipients, followed by Fluorescent in situ hybridization visualization of probes hybridizing to the Y chromosome.[28] Evidently, this cannot be applied for self-grafting, which is an area of considerable interest in regenerative medicine. We reasoned that it would be useful to develop a dye and DNA-free method for long-term ADSC monitoring that relies on expression of protein biomarkers associated with the injected cell population. Injection of MACD into rabbit larynges produces a stable ellipsoid that is visible by H&E staining at 12 weeks postinjection. This facilitates collection of tissue sections following from slides that were previously subjected to routine protocols for fixation, paraffin embedding, and sectioning. A limited number of slides (10–15) contained sufficient material to generate reproducible proteomics data in two independent runs. In the MACD+ADSC group, we identified CD9, CD44, fibronectin, and vimentin, which were previously recognized as ADSC biomarkers.[29] Nevertheless, expression of CD73, CD90, and CD105 human ADSC surface markers was not observed in rabbit ADSCs. This may be a measure of distinction between rabbit and human ADSCs because mass spectrometry analysis of freshly harvested rabbit ADSCs also did not demonstrate CD73, CD90, or CD105 expression (results not shown).

One concern framed by studies with immortalized cells is that the implant will proliferate unrestrained and ultimately interfere with organ function. Thus, if the MACD+ADSC implant continues to grow, it could distort VF tissue, narrow the glottis airway, and impair rather than improve the voice. Our previous works suggests that this may not be the case. After 4 weeks, the mean volume of tissue remaining in the MACD+ADSC group was 295 ± 102 mm,^{3} whereas at 12 weeks it was 235 ± 35 mm,^{3} representing a $\sim$20% tissue loss.[30] Thus, the volume of the implanted biomaterial is decreasing rather than increasing. Results from the literature also seem to bear out the general conclusion that the number of cells remaining at the site of implantation decreases as a function of time.[31] To extend stem cell life inside the implant, cells have been modified to overexpress a variety of growth-/proliferation-stimulating factors, such as Akt, integrin-linked kinase, Hsp-20, sFRP2, survivin, GSK-3β, ERBB4, SDF-1, and the microRNA miR-1.[23]

CONCLUSION

Future work will address the utility of biomarkers identified here to examine the full time-course of bioreagent resorption. We will also employ strategies to extend ADSC survival, such as incorporation of a modified type-2 like collagen hydrogel to extend cell life.

In conclusion, a set of biomarkers was identified by mass spectrometry analysis of fixed tissue sections of rabbit larynges at 12 weeks post-IML. These biomarkers include known ADSC surface markers, as well as atypical histones H3.3, H2A.V, and H2A.Z, which are associated with a proliferative cellular activity.

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